

	$f(x)$	$f'(x)$
	x^n	nx^{n-1}
	e^x	e^x
	a	0
	x	1
+ - + - ↑ descending	$\sin(x)$	$\cos(x)$
	$\cos(x)$	$-\sin(x)$
	$\tan(x)$	$\sec^2(x)$
	$\cot(x)$	$-\csc^2(x)$
+ - ↑ ascending	$\sec(x)$	$\sec(x)\tan(x)$
	$\csc(x)$	$-\csc(x)\cot(x)$
	$\arcsin$	$\frac{1}{\sqrt{1-x^2}}$
"↓" arc arc arc	$\cos^{-1}(x)$	$\frac{-1}{\sqrt{1-x^2}}$
	$\tan^{-1}(x)$	$\frac{1}{1+x^2}$
	$\cot^{-1}(x)$	$\frac{-1}{1+x^2}$
	$\sec^{-1}(x)$	$\frac{1}{ x \sqrt{x^2-1}}$
	$\csc^{-1}(x)$	$\frac{-1}{ x \sqrt{x^2-1}}$
	$\ln(x)$	$\frac{1}{x}$
	a^x	$a^x(\ln a)$

Product Rule:

$$\frac{d}{dx} [f(x)g(x)] = f(x)g'(x) + g(x)f'(x),$$

Quotient Rule:

$$\frac{d}{dx} \left[\frac{f(x)}{g(x)} \right] = \frac{g(x)f'(x) - f(x)g'(x)}{[g(x)]^2}.$$

Exponent Laws:

$$\begin{aligned} x^m \cdot x^n &= x^{m+n} \\ \frac{x^m}{x^n} &= x^{m-n} \\ (x^m)^n &= x^{mn} \\ (xy)^m &= x^m y^m \\ \left(\frac{x}{y}\right)^n &= \frac{x^n}{y^n} \\ x^{-n} &= \frac{1}{x^n} \\ \left(\frac{x}{y}\right)^{-n} &= \left(\frac{y}{x}\right)^n \end{aligned}$$

Dot Product:

$$\vec{a} \cdot \vec{b} = ab \cos(\theta)$$

Equation for a Line

Point Slope Form:

$$y - y_1 = m(x - x_1)$$

$$x^0 = 1$$

Chain Rule:

If $g(x)$ is differentiable at the point x and $f(x)$ is differentiable at the point $g(x)$, then $f \circ g$ is differentiable at x . Furthermore, let $y = f(g(x))$ and $u = g(x)$, then

Chain Rule ("derivative of the outside times the derivative of the inside"):

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} \longrightarrow \frac{d}{dx} f(g(x)) = f'(g(x)) * g'(x) \quad \text{ex: } \frac{d}{dx} \sin(x^2) = 2x \cos(x^2)$$

Integration by Parts:

Pneumonic: "ultraviolet vudu." Find **u** using "LogPoET" (choose **u** in this order: Logarithmic (ex: $\ln(x)$), Polynomial (ex: $2x^2+x$), Exponential (ex: e^x), Trigonometric (ex: $\sin$ & $\cos$)). The piece left over is **dv**. Differentiate **u** and integrate **dv** and then plug and chug.

$$\int u dv = uv - \int v du$$

Integration by Parts Example $\int u dv = uv - \int v du$

$$\int_0^{\infty} x e^{-sx} dx \quad (\text{polynomial is higher than exponential, so choose } u = x)$$

$$u = x$$

$$dv = e^{-sx} dx$$

$$du = 1 dx$$

$$v = -\frac{1}{s} e^{-sx}$$

$$\begin{aligned} \rightarrow \int_0^{\infty} x e^{-sx} dx &= -\frac{1}{s} x e^{-sx} \Big|_0^{\infty} + \frac{1}{s} \int_0^{\infty} e^{-sx} dx \\ &= 0 \text{ (by L'Hopital's Rule)} - \frac{1}{s^2} e^{-sx} \Big|_0^{\infty} = \frac{1}{s^2} \end{aligned}$$